

Motion planning

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1. Closed tracks

In this section we consider the problem of constructing a cubic spline to describe a 2D trajectory based on a sequence of N points $\{(X_i, Y_i)\}_{i=0, \dots, N-1}$. Here we choose by convention that the last and first point are not the same.

1.1. Spline fitting

We will create two functions $X(s)$ and $Y(s)$ of the track progress s as 1D cubic splines, i.e. $\mathcal{C}^2$ functions that are piecewise cubic polynomials. We approximate the path progress between two consecutive point (X_i, Y_i) and (X_{i+1}, Y_{i+1}) by the euclidean distance between two consecutive points:

$$\Delta s_i = \sqrt{(X_{i+1} - X_i)^2 + (Y_{i+1} - Y_i)^2} \text{ for } i = 0, \dots, N-2 \text{ and } \Delta s_{N-1} = \sqrt{(X_0 - X_{N-1})^2 + (Y_0 - Y_{N-1})^2},$$

so that we can define the global path progress for each point as

$$s_0 = 0 \text{ and } s_j = \sum_{j=0}^{i-1} \Delta s_j \text{ for } i = 1, \dots, N.$$

Note that the formula above includes the definition of the total path length s_N .

We will define the spline intervals as $[s_i, s_{i+1}]$ for $i = 1, \dots, N-2$ and $[s_{N-1}, s_0]$ (to define the N -th interval closing the spline).

Then we write $X(s)$ and $Y(s)$ as

$$X(s) = \varphi^X\left(\frac{s - s_i}{\Delta s_i}\right), Y(s) = \varphi^Y\left(\frac{s - s_i}{\Delta s_i}\right) \text{ for } s_i \leq s \leq s_{i+1} \text{ and } i = 0, \dots, N,$$

where the functions φ_i^X and φ_i^Y are the cubic polynomials on each interval:

$$\varphi_i^{X/Y}(r) = a_i^{X/Y} + b_i^{X/Y} r + c_i^{X/Y} r^2 + d_i^{X/Y} r^3 \text{ for } 0 \leq r \leq 1$$

To enforce the smoothness constraints at the junctions between the intervals, we impose the following constraints (we write them for $X(s)$ only for simplicity but the same translates to $Y(s)$):

- $X(s_i^-) = X(s_i^+)$ which translates to

$$\begin{aligned} \varphi_{i-1}^X\left(\frac{s_i - s_{i-1}}{\Delta s_{i-1}}\right) &= \varphi_i^X\left(\frac{s_i - s_i}{\Delta s_i}\right) \iff \varphi_{i-1}^X(1) = \varphi_i^X(0) \\ &\iff a_{i-1}^X + b_{i-1}^X + c_{i-1}^X + d_{i-1}^X = a_i^X \\ &\iff a_{i-1}^X + b_{i-1}^X + c_{i-1}^X + d_{i-1}^X - a_i^X = 0 \end{aligned}$$

- $X'(s_i^-) = X'(s_i^+)$ which translates to

$$\begin{aligned} \varphi_{i-1}^{X'}\left(\frac{s_i - s_{i-1}}{\Delta s_{i-1}}\right) \frac{1}{\Delta s_{i-1}} &= \varphi_i^{X'}\left(\frac{s_i - s_i}{\Delta s_i}\right) \frac{1}{\Delta s_i} \iff \frac{b_{i-1}^X + 2c_{i-1}^X + 3d_{i-1}^X}{\Delta s_{i-1}} = \frac{b_i^X}{\Delta s_i} \\ &\iff b_{i-1}^X + 2c_{i-1}^X + 3d_{i-1}^X - \rho_{i-1} b_i^X = 0 \end{aligned}$$

where we defined $\rho_i = \frac{\Delta s_{i-1}}{\Delta s_i}$.

- $X''(s_i^-) = X''(s_i^+)$ which translates to

$$\begin{aligned}\varphi_{i-1}^X''\left(\frac{s_i-s_{i-1}}{\Delta s_{i-1}}\right)\frac{1}{(\Delta s_{i-1})^2} &= \varphi_i^X''\left(\frac{s_i-s_i}{\Delta s_i}\right)\frac{1}{(\Delta s_{i-1})^2} \iff \frac{2c_{i-1}^X+6d_{i-1}^X}{(\Delta s_{i-1})^2} = \frac{2c_i^X}{(\Delta s_i)^2} \\ &\iff 2c_{i-1}^X+6d_{i-1}^X-2\rho_{i-1}^2c_i^X=0\end{aligned}$$

If we concatenate all the coefficients $a_i^X, b_i^X, c_i^X, d_i^X$ in a single vector p we can write the following linear system of equations:

$$\begin{pmatrix} 1 & 1 & 1 & 1 & -1 & & & & & & \\ & 1 & 2 & 3 & & -\rho_0 & & & & & \\ & & 2 & 6 & & & -\rho_0^2 & & & & \\ & & & & 1 & 1 & 1 & 1 & -1 & & \\ & & & & & 1 & 2 & 3 & & -\rho_0 & \\ & & & & & & 2 & 6 & & & -\rho_0^2 \\ & & & & & & & & \ddots & & \\ & & & & & & & & & \ddots & \ddots \end{pmatrix} p = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

1,2,4

1.2. Curvature minimization

2. Open tracks